EXPERIMENT: 02

Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
   char text[100];
   char key[26] = "QWERTYUIOPASDFGHJKLZXCVBNM";
   int i;
   printf("Enter the text: ");
   fgets(text, sizeof(text), stdin);
   for(i = 0; text[i] != '\0'; i++)
   {
      if(text[i] >= 'A' && text[i] <= 'Z')
         text[i] = key[text[i] - 'A'];
      else if(text[i] >= 'a' && text[i] <= 'z')
         text[i] = tolower(key[text[i] - 'a']);
   }
   printf("Encrypted text: %s", text);
   return 0;
}
```

OUTPUT:

Programiz
C Online Compiler

Programiz PRO ›

main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
    char text[100];
    char key[26] = "QWERTYUIOPASDFGHJKLZXCVBNM";
    int i;

    printf("Enter the text: ");
    fgets(text, sizeof(text), stdin);

    for(i = 0; text[i] != '\0'; i++)
    {
        if(text[i] >= 'A' && text[i] <= 'Z')
            text[i] = key[text[i] - 'A'];

        else if(text[i] >= 'a' && text[i] <= 'z')
            text[i] = tolower(key[text[i] - 'a']);
    }
```

Output

```
Enter the text: HELLO WORLD
Encrypted text: ITSSG VGKSR
```

30°C Clear
Search
ENG IN
21:33
05-10-2026